

#1C: therohan karthikeya

MY ENIGMA NUMBER:

$$E = \underline{2,285,304}$$

1C: C-FOURCHOCOLATE

MY N:

$$N = 30 + (\text{Remainder of } (E \div 70))$$

$$N = 30 + 14 = 44$$

Q1) GUESS:

(44, 39) would take more steps

Q2) EXAMPLE:

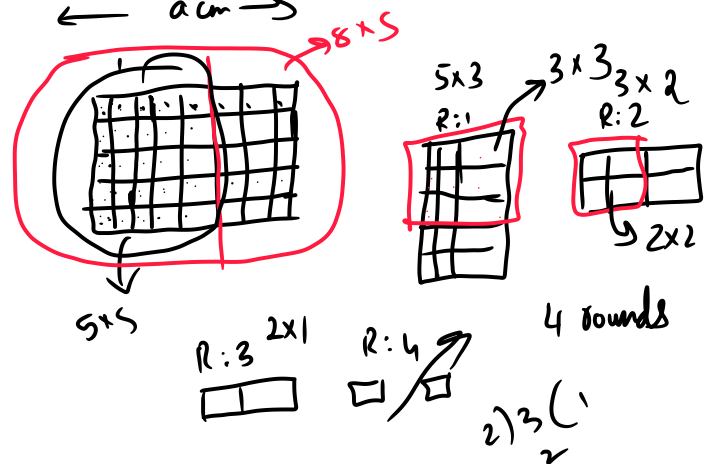
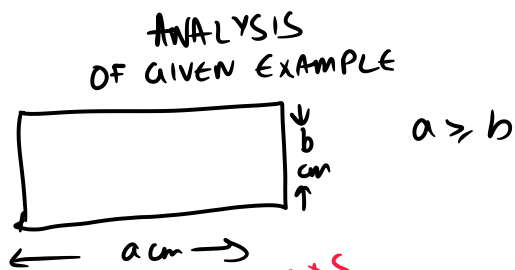
By using division algorithm $a = bq + r$, we don't need to draw any rectangles.

$$\text{for ex: } (21, 13) \Rightarrow 21 = 13 \times 1 + 8$$

$$r = 8$$

$\therefore$ The remaining chocolate length is 13×8

$\hookrightarrow$ a = longer side
 b = shorter side
 q = quotient
 r = longer side - shorter side (leftover length)



PAIR	STEPS PROCESS	NO. OF STEPS
(10, 6)	$(6, 4) \rightarrow (4, 2) \rightarrow (2, 2) \rightarrow (2, 0)$	4
(13, 8)	$(8, 5) \rightarrow (5, 3) \rightarrow (3, 2) \rightarrow (2, 1) \rightarrow (1, 1) \rightarrow (1, 0)$	6
(16, 10)	$(10, 6) \rightarrow (6, 4) \rightarrow (4, 2) \rightarrow (2, 2) \rightarrow (2, 0)$	5
(21, 13)	$(13, 8) \rightarrow (8, 5) \rightarrow (5, 3) \rightarrow (3, 2) \rightarrow (2, 1) \rightarrow (1, 1) \rightarrow (1, 0)$	7
(34, 21)	$(21, 13) \rightarrow (13, 8) \rightarrow (8, 5) \rightarrow (5, 3) \rightarrow (3, 2) \rightarrow (2, 1) \rightarrow (1, 1) \rightarrow (1, 0)$	8

Q3) The pair (34, 21) is the slowest step

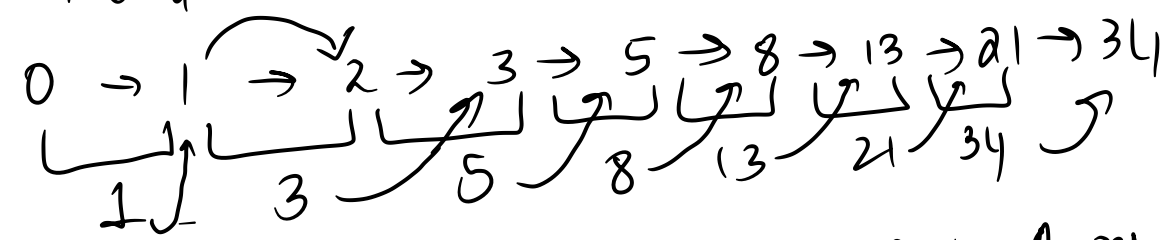
$$\begin{aligned} 34 &= 21 + 13 \\ 21 &= 13 + 8 \\ 13 &= 8 + 5 \\ 8 &= 5 + 3 \\ 5 &= 3 + 2 \end{aligned}$$

$$\begin{aligned} 3 &= 2 + 1 \\ 2 &= 1 + 1 \\ 1 &= 0 + 1 \end{aligned}$$

} 8 steps

EXPLANATION:

The slowest step occurs on (a, b) . Here (a, b) is $(34, 21)$. As I carefully observed the splitting process.



This pattern is known as Fibonacci Series. As per my observation, the consecutive pattern pairs became the slow step.

Therefore, consecutive fibonacci numbers takes the most step.

MY OBSERVATION:

The form $a = bq + r$, the q is always 1 for fibonacci series. As our goal is to take as many steps as possible, we want the remainder to be as large as possible while allowing the euclid algorithm to go slowly.